Suprabhat Kumar

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Technical Skills

Languages & Frameworks: Python, FastAPI, C++, C#, .NET Core, SQL, Bash, Go (Golang)
Cloud & Distributed Systems: Distributed Caching (Redis), Etcd, PostgreSQL, Kubernetes (AKS), Docker, REST APIs, High Availability, Concurrency, Fault Tolerance, Event-Driven Architectures
Infrastructure, Security & Windows Platform: Azure Cloud, Azure Key Vault (Secrets Store CSI Driver, Managed Identity), CI/CD, Azure DevOps (ADO) Pipelines, Windows Internals, WinUI 3, Win32/WinRT APIs, Comctl32, System Telemetry
AI Systems & Developer Workflows: Azure AI Foundry, LLM Orchestration, RAG Pipelines, PyTorch, Computer Vision, Cursor & GitHub Copilot (MCP server integrations, prompt guardrails, agentic planning workflows, rubber-duck debugging)

Experience

Microsoft **Jan 2022 – Present**
Software Engineer – Azure Cloud Services & Windows OS *Noida, India*

- **Distributed Redis Infrastructure & Data Tier:** Architected cross-region **Redis Geo-Redundancy** using open-source Redis to replicate state asynchronously across global cloud regions, serving a high-read enterprise workload of **10M+ daily reads** with sub-millisecond latency.
- Enhanced **Etcd** cluster integration by implementing dynamic leader election, distributed coordination, and resilient service discovery to guarantee automated zero-downtime failover.
- Deployed containerized **PostgreSQL** via Docker with high-availability replication topologies for multi-tenant persistent data and cross-service transactions, sustaining millions of daily operations with write-optimized indexing.
- Replaced hardcoded credentials and static keys by integrating **Azure Key Vault Secrets Store CSI Driver** with User-Assigned Managed Identity on AKS, establishing unified Azure CLI workflows across local development and cloud runtimes.
- **Autonomous Remediation Platform:** Engineered an event-driven automation platform in **FastAPI** and **ADO Pipelines** continuously auditing **100+ production repositories**, autonomously identifying and patching vulnerable dependencies and security drift.
- **AI-Powered Systems Verification (CraftGuard):** Designed an asynchronous validation pipeline combining **Azure AI Foundry** (LLM reasoning over RAG) and computer vision heuristics to automate WCAG/WinUI 3 design review, cutting manual audits by **40–60%**; implemented with a **FastAPI** backend and a native **C#** Windows client.
- **Windows Platform Modernization:** Modernized critical legacy OS shell surfaces (File Properties, Copy Dialog, Confirmation Dialog) from legacy **Comctl32** to **WinUI 3**, delivering native dark mode, touchscreen accessibility, and volume device routing to billions of users.
- Engineered end-to-end production telemetry pipelines in **C#/.NET** and native **C++ (Win32/WinRT)** across staged deployment rings, tracking reliability, failure rates, and performance regressions for millions of active endpoints.
- **Engineering Velocity & AI Workflows:** Standardized team developer practices using **Cursor** and **GitHub Copilot** with custom **MCP server** tool integrations, structured plan-mode prompt guardrails, and rubber-duck problem decomposition.

BlogVault **Jan 2021 – Dec 2021**
Software Engineer – Backend & Distributed Storage *Bengaluru, India*

- Engineered backend microservices and asset optimization engines in **Ruby on Rails** and **PHP/MySQL**, automating minification, caching, and compression across thousands of high-traffic sites, boosting page loading speeds by **25%**.
- Designed an edge-caching and asset aggregation engine reducing origin server load by **35%**; optimized concurrent database queries and Redis caching to eliminate database locking across continuous multi-tenant backup pipelines.

IIT Patna **May 2020 – Jul 2020**
Software Engineering Intern – Conversational AI *Patna, India*

- Developed a task-oriented conversational dialogue engine using **PyTorch** to parse shopping intents and slot constraints, executing dynamic database querying hooks to fetch inventory and synthesize context-grounded responses.

Education

National Institute of Technology, Patna **2017 – 2021**
*Bachelor of Technology (B.Tech) in Computer Science and Engineering – **CGPA: 8.66 / 10.0*** *Patna, India*

Achievements & Honors

Competitive Programming: Peak Rating **1700+** on Codeforces (*suprabhatkr*); solved 800+ algorithmic problems.
Production Ownership: Owned Tier-1 Windows feature release cycles from native development through staged retail rollout rings at Microsoft.
Smart India Hackathon (SIH) 2020: National Finalist among 10,000+ engineering teams for building resilient offline-first mobile sync architectures.